Individual Learning Contract - SEDE Studios

This is your Individual Learning Contract. Here you provide:

- An outline of your team's product
- A list of the technical and non-technical knowledge and skills required by the team to develop and deliver the product.
- The extent to which your team members currently have the required skills and knowledge.
- Your learning goals.

You will detail two technical and two non-technical learning goals which you are committing to achieve over the session. At least three of these goals should support your team's development of the team product so your understanding of the product requirements and the team's current skills and knowledge should guide your choice of goals.

This is the beginning of the session and you have only started the process of forming a team and choosing and developing your product. Your understanding of the product needs and the team member's capabilities may change over the session, so this contract reflects the understanding you have gained through your initial investigations and the discussions you have had with your team and product owner.

You are encouraged to re-evaluate your learning needs as you progress through the session, however we want to see that you have undertaken a careful analysis of your product and your team to begin the process.

The ILC Agreement is worth 10% of your final mark so it should reflect about 12-15 hours of effort on your part. You should spend those hours talking to your team and your product owner and identifying suitable learning goals, and then identifying and researching activities and resources that will allow you to achieve those goals.

You will need to use the free Adobe Acrobat Reader to complete this form.

About You

Student ID	24493999
Full Name	Nathan Nguyen
Subject Enrolled	Professional Studio A
Your Major	Electrical and Electronic Engineering

About Your Product

Before completing this section, you should discuss the product with your product owner and the rest of the team and develop a common understanding of the product. Provide below a brief outline in your own words of the common understanding you have arrived at.

We understand that all team members' descriptions will be similar.

Typical FMCW (Frequency Modulated Continuous Wave) Radar systems measure distance and speed of objects. Our team is pushing the boundary of typical radar applications: we are using it to identify and distinguish between different materials based on their electromagnetic properties, particularly dialectric characteristics.

The scope of this project is wide, including review of existing radar detection research to design and development of our own experimental radar platform to testing and validation in the lab. The final product should be added in prototype that proves FMCW radar can be used as a material identification tool.

A key focus is reliability. The system needs to produce constant high precision results in a controlled environment, so we can properly benchmark its performance and understand how accurately it can classify different materials.

About Your Team

Please reconfirm for us the names of all of your team members below, and which studio and which major they are completing. Include yourself.

e.g. Josephine Wu, Applications Studio B, Electrical and Electronic Engineering

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Aung Pyay - professional studio B, electrical and electronic engineering
Nathan Nguyen - professional studio B, electrical and electronic engineering
Don Weerakoon - professional studio A, electronic engineering
Joshua Dyer - professional studio B electrical engineering
Ilana Maryasin -
Senanayaka Amarasinghege - professional studio A, electronic engineering

About Your Product Needs

Before completing this section, you should have a discussion with your team about the critical skills and knowledge required to develop and deliver your product. These should include both technical and non-technical skills and knowledge.

Technical and Non-Technical Skills and Knowledge

It is often clear what the technical needs of a product are. For example, you may need to develop a printed circuit board that interfaces to a single board computer connected to sensors to read data, process those inputs then feed them across some communications channel to a computer that runs an algorithm to make sense of that data and present it in some visualisation to a user. Maybe you need to design an enclosure to protect that PCB and the single board computer. All of those things are technical needs, and most engineers can do a pretty good job of identifying the required skills and knowledge to meet those needs.

But what are the non-technical needs?

Does your product solve a real problem that enough people care about?
You must be able to develop a clear understanding of the market need your product will meet.

Can you convince your boss or a venture capital provider or that key engineer, to help you create it?
You need to be able to communicate your ideas effectively to people who may not understand or care about the technical aspects of your product.

Can you deliver the product on time and to specification?
To successfully create your product, you need to understand how to organise and energise a team.

Will your product meet the relevant standards or regulations? Does it require a license to an existing patent? Should you seek a patent to protect some novel part of it.
You need to understand the intellectual property, regulatory and standards relevant to your product.

Even if you are not intending to take a more entrepreneurial path in your career, you still need non-technical skills. If you apply for a job and there are 300 applicants, often the skills that will distinguish you from the other 299 applicants are non-technical skills.

Technical Product Needs

These should include technical skills and knowledge in such areas as:

• Electronic circuit design • PCB layout • Single board computers • Sensors • Programming • Version control • Data analysis • Databases • Electronic communications • Website frontend development	• Website backend development • Cloud computing • Data visualisation • Machine learning • Image processing • Signal processing • 3D modelling • 3D printing • Raspberry Pi • Arduino

Be specific where possible. So, if your team has decided to use a particular tool or language, specify that.

List the four to six most critical technical skills and knowledge required to develop your product. All team members can put the same list here.

Num	Most Critical Technical Product Needs
T1	Signal Processing - Ability to analyse FMCW radar signals to differentiate materials properties.
T2	Electronic Communications - Ability to understand transmission, reception, and reflection of electromagnetic signals in a radar system.
T3	Data Analysis - Ability to interpret results and compare data from different material properties
T4	Programming - Ability to use Python or MATLAB to process radar data and support testing.
T5	Electronic circuit design - Ability to understand radar hardware platform
T6	Sensors - Ability to configure and validate the radar system as a sensing platform for material detection.

Non-Technical Product Needs

Non-technical Product needs should include skills and knowledge in such areas as:

• Technical report writing • Presentation skills • Communication skills • Record keeping skills • LabArchives • Team forming • Team leadership • Team coordination • Time management	• Project management • Product development paradigms • Domain knowledge • Market understanding • Requirements analysis • Use case modelling • Intellectual Property • Standards • Regulatory constraints

Again, be specific where possible. So, if your team has decided to use to use a particular product development paradigm such as Agile Development, specify that. If your team is planning to use Trello for team coordination, specify that.

List the four to six most critical non-technical skills and knowledge required to develop your product. All team members can put the same list here.

Num	Critical Non-Technical Product Needs
NT1	Domain Knowledge - Understand radar detection systems and material sensing concepts.
NT2	Requirement analysis - Define project objectives and evaluate whether they are achieved.
NT3	Technical Report Writing - Document, research, experiment, validate findings clearly
NT4	Team Coordination - Organise team tasks and integrate contribution
NT5	Presentation Skills - Communicate project progress and outcome effectively
NT6	Time Management - Complete objectives, deliverables and milestones on time, without missing deadlines.

About You Team's Skills and Knowledge

To complete this section, you will need to have listed as thoroughly as you can, the technical and non-technical skills and knowledge required to develop your product. Here you will identify the level of competence of each team member (0 = none, 4=excellent) in the most critical product needs. All team members can put the same table here.

Product Needs	Technical						Non-Technical					
Team Member	T1	T2	T3	T4	T5	T6	NT1	NT2	NT3	NT4	NT5	NT6
Aung P	3	3	3	3	3	3	2	4	4	4	4	4
Don W	1	1	2	1	2	2	2	4	2	1	1	1
Joshua D	3	3	2	3	3	3	3	4	3	2	3	3
Adrita A	1	1	1	2	1	3	1	1	2	3	3	2
Senanayaka A	3	1	4	1	2	3	1	2	1	3	1	2
Nathan N	1	3	2	2	2	3	3	4	3	3	3	4

Write a brief reflection on the team's skills and knowledge.

Unlike the table above, this should be **your own reflection** (not the same as your team members).

Looking at the team's skill ratings, there are clear strengths and gaps worth noting. Technically, the team has solid coverage in signal processing (T1) and sensors (T6), with Aung, Joshua, and Senanayaka leading in those areas. However, skills like electronic communications (T2) and programming (T4) are weaker for some members — particularly Don and Senanayaka — which could pose a risk during the implementation phase of the radar system.
On the non-technical side, the team is generally strong in requirements analysis (NT2) and time management (NT6), which is encouraging for project delivery. The main gap is domain knowledge (NT1), where several members rated themselves low, which makes sense given that FMCW radar material sensing is a fairly niche area that none of us have directly worked with before.

Your Learning Goals

From your analysis of the team's skills and knowledge, identify at least two technical and two non-technical learning goals that you can pursue over the session. At least three of these goals should directly support your team's product development. You can choose one goal that is not related to the product development but broadly supports your professional development as an engineer and is related to your major or sub-major.

It may be that you will find that you need to change your learning goals or the activities or resources that you identify here as you progress through the session. That is OK, just be sure to document and justify the new goal in the record you keep of your learning activities.

Learn Different Things; Close the Gaps

In selecting your learning goals, you should coordinate with your team members to ensure that the team achieves a good coverage of the gaps in the team's capabilities against the product needs, as identified in your analysis above.

In general, we expect each member of the team to choose different learning goals and activities. While there may be some overlap, e.g. two team members might learn the same topic to help support each other in the product development, when comparing your 4 learning goals with those of your team members, no two team members should have all 4 goals in common. One or two at most is acceptable.

Teach the Team

If you ask any lecturer, they will tell you that teaching a subject to someone else is the most effective way to test and consolidate your knowledge of the subject. We ask all Fundamentals and Applications students to nominate one of their learning activities which they will use to develop a brief (20 minute) "Teach the team" session to transfer some of their learnings to their team.

Typically you will complete this in weeks 7-9 (or stuvac). i.e. you need to have learned the topic yourself first before you can teach it, but you also need to teach it before the final submission of your ILC Delivery.

You later need to provide evidence of having completed your Teach-the-team activity in your ILC Delivery records.

Professional students are not required to do this (but may choose to).

SMART Goals

In choosing your learning goals you must use a SMART approach. This means that the goals should be:

- Specific
- Measurable
- Achievable
- Relevant
- Timebound

This form is structured to help you to achieve that, but you can learn more about the broader use of SMART goals in this video

https://www.youtube.com/watch?v=qxQI5tyRUes

For the purposes of the ILC this means:

Specific – clearly identify what skills or knowledge you will gain. You must also identify the specific resource(s) you intend to use. This means a book or a paper (provide a reference) or a specific online course or online video tutorial (provide a URL). Just “youtube.com” is **not** sufficient.

Measurable – To show that you have achieved this goal, you will be required to submit your ILC notes regularly from week 4, and by week 10 provide a brief reflection on what you have learned and how it has impacted the product development or your professional development. If you use an online course that provides a certificate on completion, then you should also provide that certificate in your record of your learning activities at the time you complete the course. If you are following YouTube tutorials or reading a reference book, you will need to consider how you will collect evidence (measure) the learning you have completed.

Achievable – You should ensure that you have the pre-requisite skills to undertake this learning goal and that you can complete it in the time available.

Relevant – You must identify the product need that this goal addresses (or that it addresses your development as a professional engineer).

Timebound – You should provide an estimate of the hours required to complete the activity and the date you intend to complete it. Each activity should require at least 6 hours but no more than 20 hours of work. You should map out the sequence and dates by which you plan to complete each activity. All ILC activities must be complete by the end of week 10 (but by week 8 is preferred), so you have time to apply the learning to your product development.

First Technical Learning Goal

S Specific	**What specific skills or knowledge will you gain?**	
	Python/Matlab/C++ programming used for data analysis and testing.	
	What specific resource or resources will you use? **Provide the precise URLs below to each resource.**	
	freeCodeCamp – Data Analysis with Python https://www.freecodecamp.org/learn/data-analysis-with-python/ MathWorks – Signal Processing Onramp (free) https://matlabacademy.mathworks.com/details/signal-processing-onramp/signalprocessing	
M Measurable	**What evidence will you collect to demonstrate your learning?**	
	Jupyter notebooks showing data cleaning and visualisation. MATLAB scripts demonstrating spectral analysis and filtering.	
A Achievable	**How does this build on your current knowledge? (e.g. previous studio)** **Or is it a brand new topic for you?**	
	I use python and MATLAB semiregularly, but never for anything this complex.	
R Relevant	**Need addressed: T1…T6 for product needs** **P for professional devel. need (at most one)**	T4
	If T1…T6, in what way will you use this knowledge in your product?	
	Extracting insights and verifying success	
	Will you use this activity to create your **"Teach the Team" session (Y/N)**	Y
T Timebound	**How many hours will this take? (at least 6 hours on each goal).** **If you are using online courses, specify how many hours each one says it requires, and briefly explain your calculation.**	
	10-12 hours	
	What date do you intend to complete the learning? **(before week 10, not all tasks due at once)**	
	Week 9	

Second Technical Learning Goal

S Specific	**What specific skills or knowledge will you gain?**	
	Sensor configuration and analysis	
	What specific resource or resources will you use? **Provide the precise URLs below to each resource.**	
	Texas Instruments – Intro to mmWave Sensing: FMCW Radars (5-part free video series)https://www.ti.com/video/5415203482001 Radar Tutorial – FMCW Radar & Dielectric Properties (free reference) https://www.radartutorial.eu/02.basics/Frequency%20Modulated%20Continuous%20Wave%20Radar.en.html	
M Measurable	**What evidence will you collect to demonstrate your learning?**	
	A written summary document comparing how different material dielectric constants affect the FMCW radar return signal, incorporating sensor calibration and validation principles applied during lab setup and testing.	
A Achievable	**How does this build on your current knowledge? (e.g. previous studio)** **Or is it a brand new topic for you?**	
	Have done sensors previously, but never on this scale	
R Relevant	**Need addressed: T1…T6 for product needs** **P for professional devel. need (at most one)**	T4
	If T1…T6, in what way will you use this knowledge in your product?	
	Sensors are essential for gathering accurate and precise data to validate and analyse.	
	Will you use this activity to create your **"Teach the Team" session (Y/N)**	N
T Timebound	**How many hours will this take? (at least 6 hours on each goal).** **If you are using online courses, specify how many hours each one says it requires, and briefly explain your calculation.**	
	10 Hours	
	What date do you intend to complete the learning? **(before week 10, not all tasks due at once)**	
	Week 9	

First Non-Technical Learning Goal

S Specific	**What specific skills or knowledge will you gain?**	
	Requirement analysis	
	What specific resource or resources will you use? **Provide the precise URLs below to each resource.**	
	Rice University – Engineering Project Management: Initiating and Planning (audit free)https://www.coursera.org/learn/initiating-planning	
M Measurable	**What evidence will you collect to demonstrate your learning?**	
	A requirements document drafted for the FMCW radar project defining objectives, success criteria, and evaluation checkpoints	
A Achievable	**How does this build on your current knowledge? (e.g. previous studio)** **Or is it a brand new topic for you?**	
	Scoped out projects previously, but measuring success against the scope is always a bit arduous	
R Relevant	**Need addressed: NT1…NT6 for product needs** **P for professional devel. need (at most one)**	NT2
	If NT1…NT6, in what way will you use this knowledge in your product?	
	This requirement analysis defines the scope of what needs to be performed and judges whether or not we succeded.	
	Will you use this activity to create your **"Teach the Team" session (Y/N)**	N
T Timebound	**How many hours will this take? (at least 6 hours on each goal).** **If you are using online courses, specify how many hours each one says it requires, and briefly explain your calculation.**	
	10 hours	
	What date do you intend to complete the learning? **(before week 10, not all tasks due at once)**	
	Week 9	

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Second Non-Technical Learning Goal

S Specific	**What specific skills or knowledge will you gain?**	
	Technical Report Writing	
	What specific resource or resources will you use? **Provide the precise URLs below to each resource.**	
	University of Sheffield – Technical Report Writing for Engineers (FutureLearn, free to audit) https://www.futurelearn.com/courses/technical-report-writing-for-engineers	
M Measurable	**What evidence will you collect to demonstrate your learning?**	
	LaTeX files dmeonstrating clear and defined structure of the report .	
A Achievable	**How does this build on your current knowledge? (e.g. previous studio)** **Or is it a brand new topic for you?**	
	Have written reports before	
R Relevant	**Need addressed: NT1…NT6 for product needs** **P for professional devel. need (at most one)**	NT4
	If NT1…NT6, in what way will you use this knowledge in your product?	
	Communicating findings and methodology is essential for replication and validation amongst peers.	
	Will you use this activity to create your **"Teach the Team" session (Y/N)**	Y
T Timebound	**How many hours will this take? (at least 6 hours on each goal).** **If you are using online courses, specify how many hours each one says it requires, and briefly explain your calculation.**	
	6 hours	
	What date do you intend to complete the learning? **(before week 10, not all tasks due at once)**	
	Week 7	